

UNIT I

BUILDING MATERIALS

- STONES
- BRICKS
- CEMENT
- CONCRETE
- STEEL



CEMENT - Introduction

- It is the product which is obtained by burning and crushing to powder with a well proportioned mixture of calcareous and argillaceous materials.
- It was introduced in 1824 by Joseph Aspdin, a brick layer of Leeds, England.
- After hardening, it resembles in colour similar to the stone quarried near Portland in England. So, it is named as Portland cement.
- It is available in different varieties and the selection of variety depends upon the condition, strength and the type of structure.

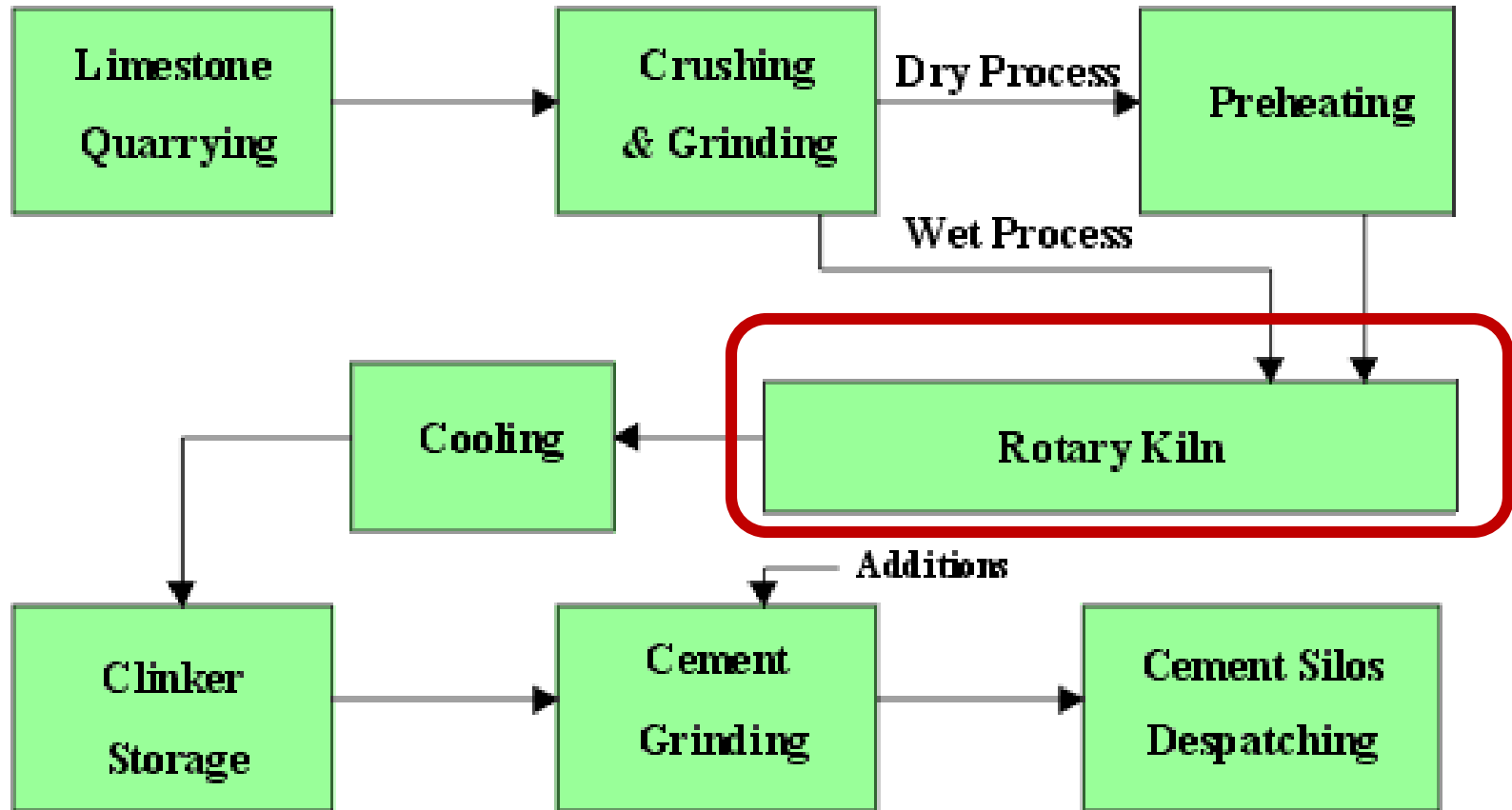
Raw Materials for Cement

- Calcareous material – Containing CaCO_3 (primary source – limestone); impurities such as iron and alumina are sometimes present
- Argillaceous material – Containing clayey matter, source of SiO_2 , Al_2O_3
- Gypsum – Added in the final stages of manufacture as a set regulator
- Ground limestone (or even fly ash) is also added to cement in varying quantities (IS permits up to 5% in OPC)

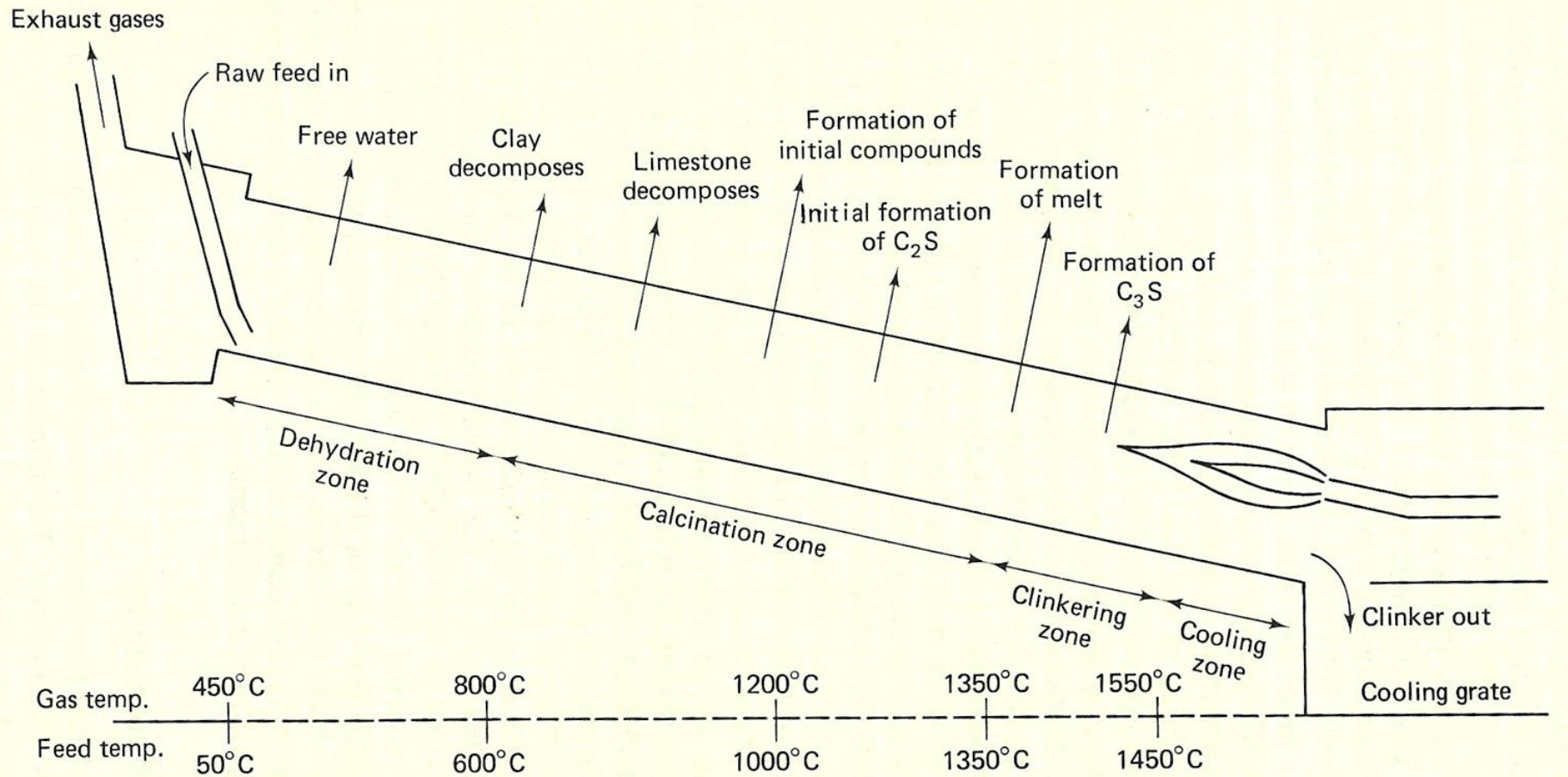
Main chemical Ingredients of Cement (Raw Stage)

Sl. No.	Ingredient	Composition	Percentage
1	Lime	CaO	60 to 67
2	Silica	SiO ₂	17 to 25
3	Iron Oxide	Fe ₂ O ₃	0.5 to 6
4	Alumina	Al ₂ O ₃	3 to 8
5	Gypsum or Calcium Sulphate	CaSO ₄	3 to 4
6	Magenisum Oxide	MgO	0.1 to 4
7	Sulphur Trioxide	SO ₃	1 to 3
8	Alkalies such as soda	NaOK ₂ O	0.1 to 1

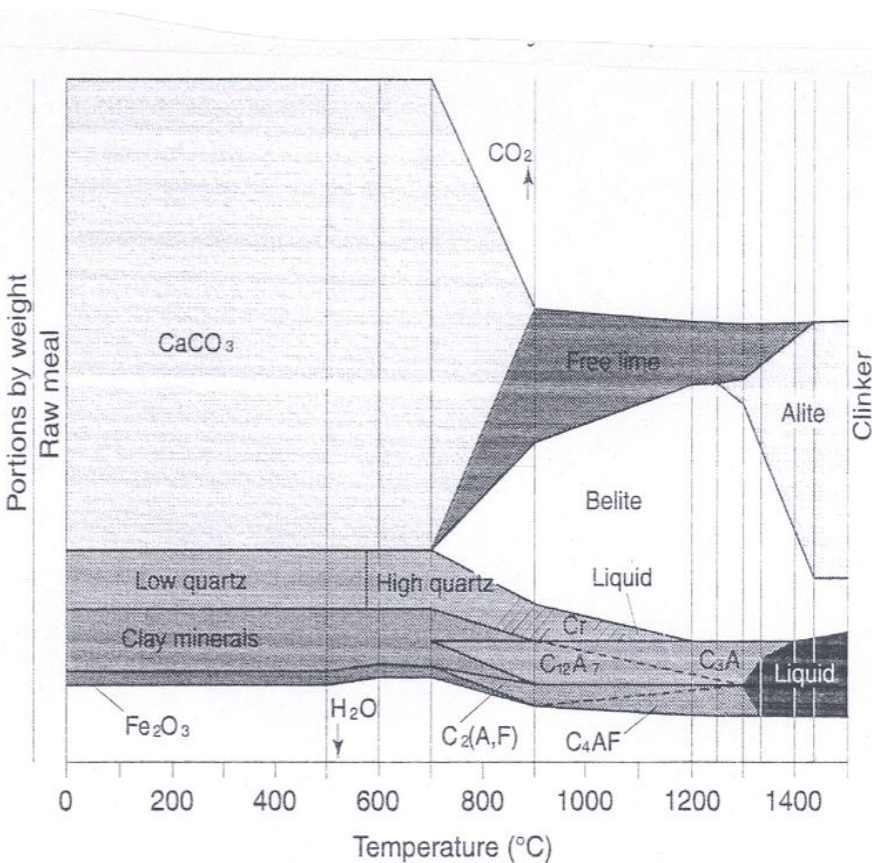
Schematic of cement manufacturing process



Kiln reactions - schematic



Kiln reactions to form different compounds



Hewlett, 2001

Up to 700 °C: activation of silicates through removal of water and changes in crystal structure

700 – 900 °C: decarbonation of CaCO_3 , initial combination of A, F, and activated silica with lime

900 – 1200 °C: Belite (C_2S) formation

> 1250 °C (more particularly, > 1300 °C): liquid phase appears and promotes the reaction between belite and free lime to form alite (C_3S)

Cooling stage: molten phase (containing C_3A and C_4AF) gets transformed to a glass; if cooling is slow, C_3A crystallizes out (causes setting problems), or alite converts to belite and free lime

Inter-grinding Clinker with Gypsum (CaSO_4)

- Final step in cement manufacture = (Clinker + Gypsum)
- Gypsum added as a set regulator (absence $\rightarrow$ flash set)
- Strict control on temperature is required
- Done in ball mills; vertical roller presses are now used for better efficiency
- Cement of required fineness produced by grinding in ball mills

Final compounds of Cement

Sl. No.	Compound	Formula	Abbreviation	Range
1	Dicalcium silicate	$2(\text{CaO}).\text{SiO}_2$	C_2S	21 to 45%
2	Tricalcium silicate	$3(\text{CaO}).\text{SiO}_2$	C_3S	25 to 50%
3	Tricalcium Aluminate	$3(\text{CaO}).\text{Al}_2\text{O}_3$	C_3A	5 to 11%
4	Tetra Calcium Aluminum Ferrite	$4(\text{CaO}).\text{Al}_2\text{O}_3$ $.\text{Fe}_2\text{O}_3$	C_4AF	9 to 14%
5	Other Constituents and Gypsum	-	-	8%

Final compounds of Cement contd.....

➤ C_2S

- It hydrates slowly.
- It hardens more slowly
- It provides ultimate strength to the cement.
- It has more resistance to chemical attack.

➤ C_3S

- It hydrates more rapidly.
- It develops early strength.
- It generates heat more rapidly and has less resistance to chemical attack.

Final compounds of Cement contd.....

➤ C_3A

- It is fast reacting with large amount of heat generation.
- It causes initial setting of cement.
- It is weak against sulphate attack.

➤ C_4AF

- It is comparatively inactive.
- It has poor cementing value.
- It is slow in reaction with small heat generation.

Hydration of Cement

- Reaction of cement with water
- Exothermic; heat released is called 'Heat of Hydration'
- Rate of heat evolution is faster if the reaction is quicker
- Heat evolved depends on reaction rate of individual compounds, and also on the clinker morphology !

Rate of Heat Evolution During Hydration

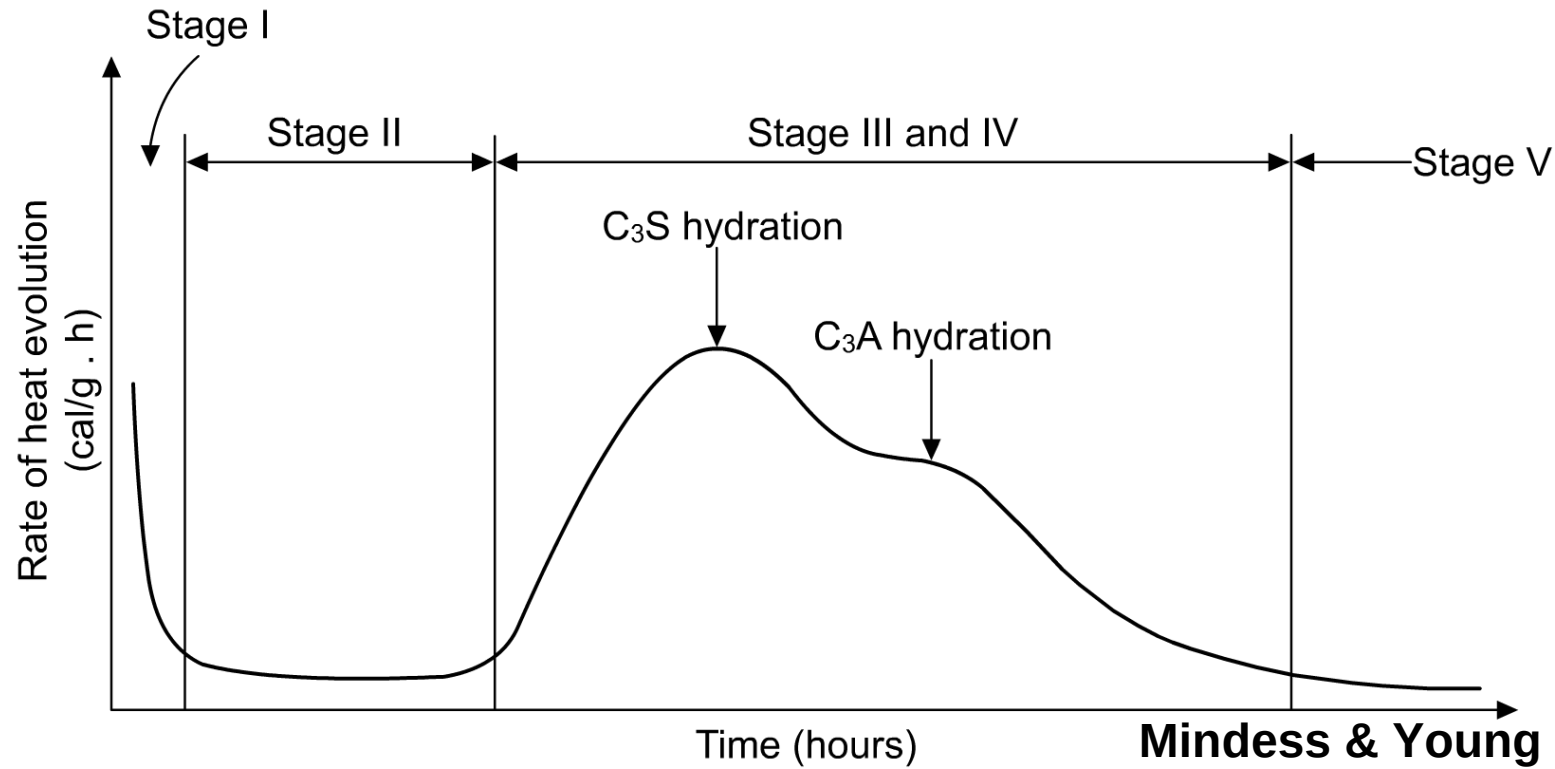
Stage I: Rapid evolution of heat, lasts about 15 minutes.

Stage II: Dormant period, lasts until initial set occurs in 2 to 4 hours

Stage III: Rapid reaction of C_3S during the acceleration period, with the peak being reached at about 8-10 hours, much after final set at 4-8 hours and hardening has begun

Stage IV: Rate of reaction slows down until steady state is reached in 12-24 hrs

Stage V: Steady state



Hydration of Cement:

Formation of Calcium Silicate Hydrate (C-S-H)



Moderate reaction rate, gives high strength and more heat liberation

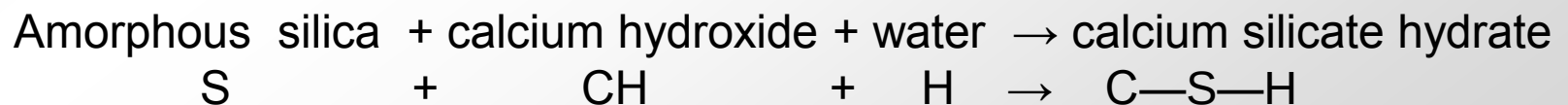


Slow reaction rate, gives low initial but high later strength, and low heat liberation

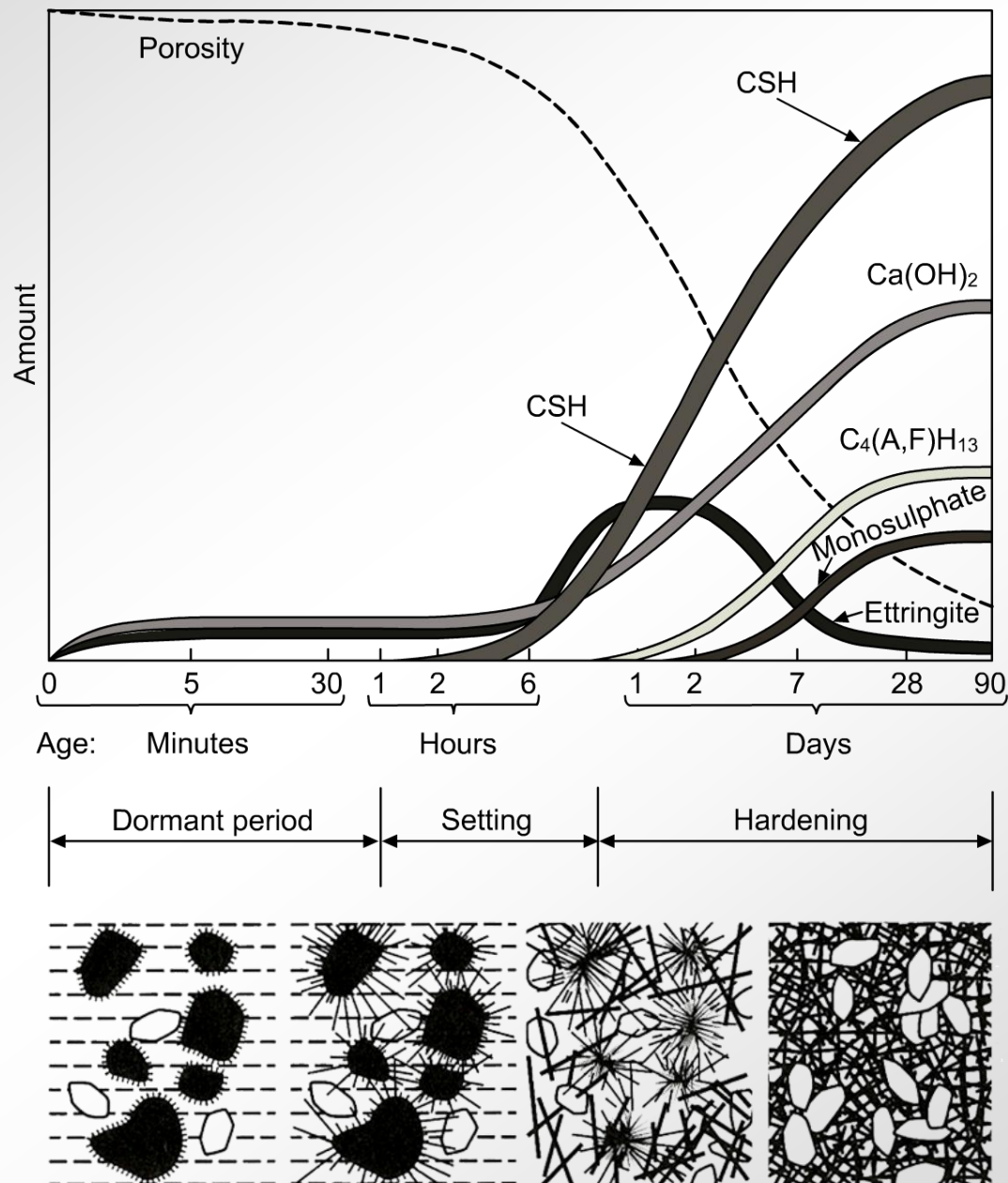


Comparison of Silica fume, Fly ash and Cement

	Silica Fume	Fly Ash	Cement
SiO ₂ Content (%)	85- 97	35 - 48	20 -25
Surface Area (m ² /kg)	17,000 - 30,000	400 - 700	300 - 500
Pozzolanic Activity (with cement, %)	120 - 210	85 - 110	n/a
Pozzolanic Activity (with lime) (MPa)	8.3 - 11.4	5.5 - 6.9	n/a

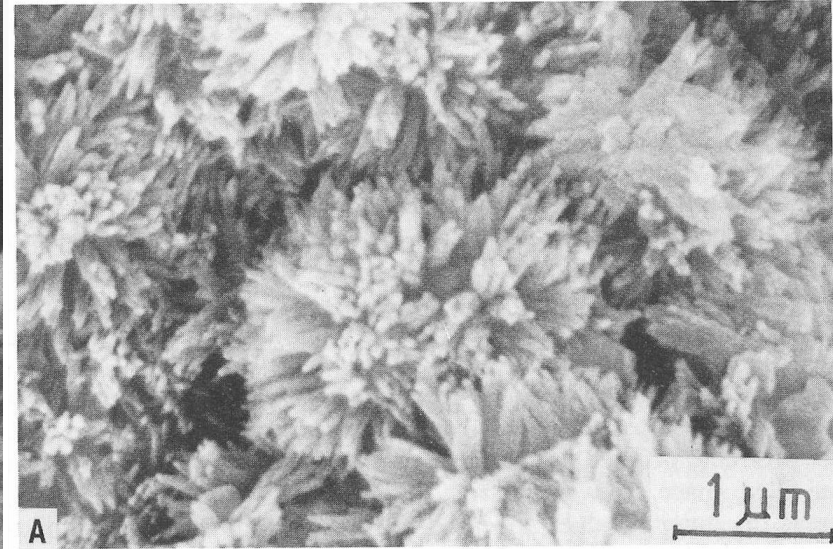
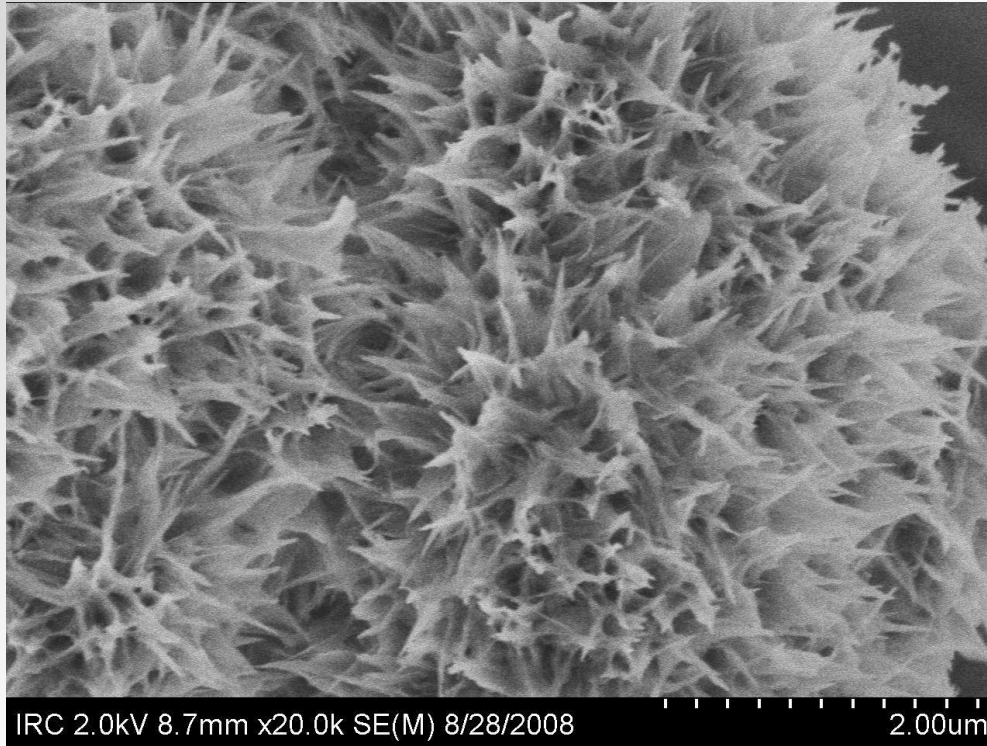


Evolution of different production by hydration



Young et al.

Scanning Electron Microscope (SEM) image of CSH



Taylor, 1968

Fineness of Cement by Blaine air permeability method

- **Fineness:**
 - It is the degree of grinding of cement.
 - The rate of reaction depends upon the fineness of grinding .
 - For accurate measurement, it is measured by surface area, air permeability method and the unit is m^2/kg of cement.
 - 200 to 300 m^2/kg of cement



Fineness of Cement – by standard sieve method

- 100 gm of cement is weighed accurately, placed on IS: sieve No. 9 (90 micron) and seived
- The residue left is weighed. This shall not exceed 10% by weight of the sample.
- This is estimated in terms of the specific surface, i.e., the surface area per unit weight.



Setting Time: Initial & Final

For initial setting



For final setting

- Setting of cement is the phenomenon by virtue of which the green cement changes into hard mass.
- The time between water is added in cement and initial setting takes place is known as Initial Setting Time.
- Initial setting is a stage in the process of hardening after which any crack that may appear will not reunite and the completion of this process is, known as final setting time
- Cement should not lose its plasticity till the various operations of mixing, transporting and placing are complete. Hence this time is generally kept not less than 30 minutes and the final setting is not more than 10 hours

Vicat's apparatus



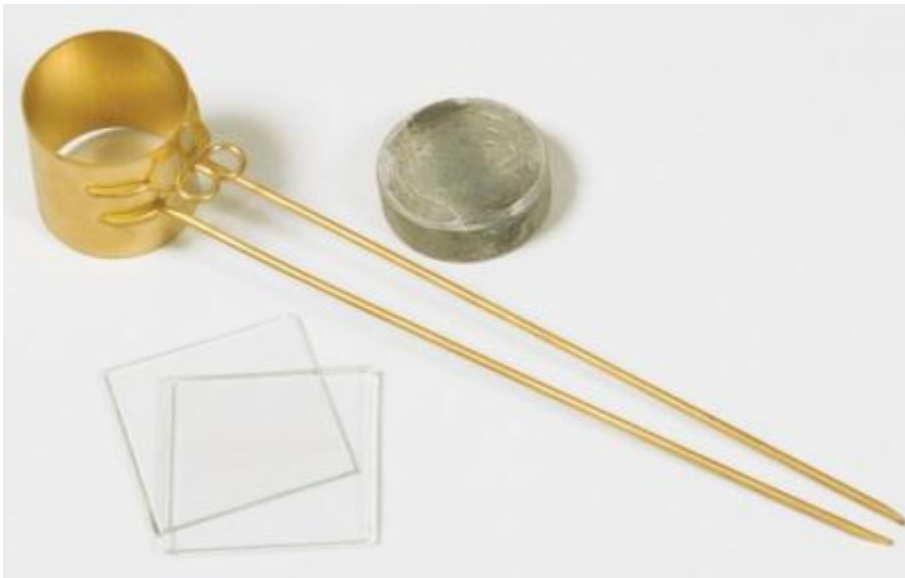
Compressive Strength:

- The quality of concrete and cement is always judged by strength and that is only by compressive strength because cement is weak in tension and for it steel reinforcement is always provided.
- For this purpose cement and standard sand are mixed in the ratio of 1:3
- Grades of cement : 33, 43, 53
(Strength (in MPa) on 28th day of curing)



Soundness (Le-Chatelier method)

- Free lime and magnesia present in cement makes the cement unsound by increasing the volume after setting.
- Expansion should not be more than 10 mm
- More lime (or calcium hydroxide) leads to cracks in concrete
- It is generally measured by Le-Chatelier method or by Autoclave method.



Field Tests on Cement:

- The colour of cement should be uniform gray with light greenish shade.
- Cement should feel smooth when touched.
- If hand is inserted in a bag of cement it should feel cool not warm.
- If a small quantity of cement is thrown in a bucket of water, it should sink and should not float on the surface.
- Cement should be free from any hard lumps.

Different Types of Cement

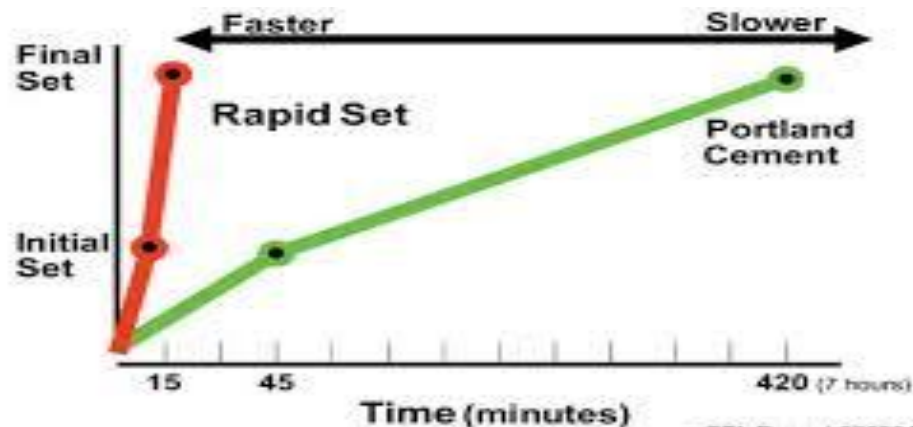
1. Ordinary Portland cement.
2. Rapid Hardening cement
3. Low Heat cement
4. Portland slag cement
5. Portland Sulphate resistant cement
6. Air entraining cement
7. White and coloured cement
8. High Alumina cement
9. Pozzolana cement
10. Oil well cement
11. Quick setting cement
12. Expanding cement
13. Hydrophobic cement

1. Ordinary Portland Cement (OPC):

- It is also known as Normal setting cement.
- It is used in road pavements, buildings, culverts, water pipes etc.
- Out of the total consumption of different types of cement 90% is OPC.

2. Rapid Hardening Cement:

- It contains less quantity of C_2S and more quantity of C_3S .
- It is generally used where high early strength is required.
- It is used by concrete product manufactures, highway pavements which are to be opened early for road traffic and in cold weather concreting due to its high heat of hydration development.



3. Low Heat Portland Cement:

- This cement is so called because it develops low heat at the time of hydration.
- It contains C_3A and; C_3S in less quantity because they develop early heat.
- It develops strength quite late and it is generally used in massive concrete structures such as dams, bridge, abutments, retaining walls etc.

4. Portland Slag Cement:

- Percentage of blast furnace slag varies from 25 to 65%.
- It is cheaper as compared to ordinary cement because waste product is used in it.
- It can also be used in massive concrete such as dams, bridges, etc.

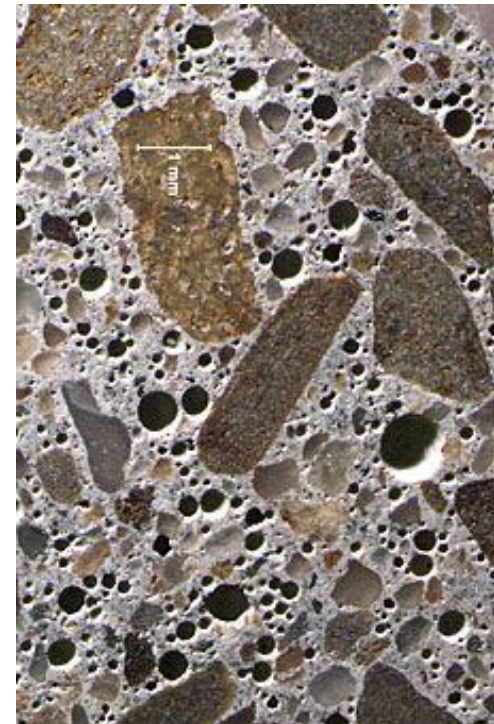
5. Sulphate Resistant Portland Cement:

- It contains very low percentage of C_3A and C_4AF .
- It is used in canal lining, construction of sewer lines where acid formation is expected.



6. Air Entraining Cement:

- It is ordinary Portland cement mixed with small quantities of air entraining materials during grinding.
- The diameter of air bubbles varies from 0.075 mm and 1.25 mm.
- On account of air bubbles the strength of cement is reduced.
- Air bubbles are permitted only up to 3 to 4 %, as these reduce 10 to 15% strength of cement.
- This cement is more plastic and workable causing less segregation and bleeding in concrete.
- It also reduces the water requirement.
- High resistance to freeze-thaw.



7. White and Coloured Cement:

- White chalk and china clay are used instead of lime stone and clay as these are having low percentage of Iron Oxide (i.e.1 %).
- Oil is used as fuel to avoid contamination of coal ash. It is 3 to 4 times costlier than ordinary cement.
- It is used for decorative floorings.
- For coloured cement, suitable pigments varying from 5 to 10% free from soluble salts are added during grinding.



8. High Alumina Cement:

- It contains 35% to 45% of aluminates, bauxite and chalk or lime stone are mixed dry and heated till they melt and on cooling they form clinkers.
- It is dark in colour and initial setting time varies from 3 to 6 hours and final setting takes place within 2 hours of the initial set.
- It gives high heat of hydration and, is also costlier than ordinary Portland cement.
- It is used in structures subjected to the action of sea water, chemical and sulphate bearing water.

9. Pozzolana Cement:

- Pozzolana is a naturally occurring material such as volcanic ash or Pumice stone or an artificial product such as burnt clay or shale containing siliceous and aluminous mineral substances.
- As per BIS 1489-1967, the proportion of pozzolana material varies from 10 to 25% by weight of cement.
- It increases the workability, reduces heat of hydration, and increases the water tightness.
- It also offers greater resistance against sulphatic action and sea water.

10. Oil Well Cement:

- As the name indicates, it is used for cementing of oil wells.
- It is used at greater depth, under high temperature and pressure.
- Iron Oxide is so adjusted that all the alumina is converted into C_4AF and proportion of C_3A is very small, increases the setting time of cement and also hardens quickly after setting.
- It protects the oil well causing from corrosion and also help in supporting the oil well casing and thus reduce the earth pressure on steel pipes.

11. Quick Setting Cement:

- It has less proportion of $CaSO_4$ (Gypsum) or a small amount of aluminum sulphate is added at the time of grinding.
- Its initial setting time is 5 minutes and final setting time is 30 minutes.
- Used in running water construction sites

12. Expanding Cement:

- As per the name, its volume increases on hardening.
- It takes about 15 days for the expansion to occur completely but the time can be controlled by curing.
- The upper limit of expansion is 1%.
- Uses to repair cracks.



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